Chapter 8. Inference in Multiple Regression II

Extra sums of squares, partial and sequential F, joint regions, simultaneous intervals, equality of regressions

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8.1 Extra sums of squares

When regressors are added to a model, SSR increases (and SSE decreases). The increase is the **extra sum of squares** due to the added regressors, and it measures their contribution *given* those already present.

For the full model $y = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \epsilon$, write

$$SS(\hat\beta_0, \ldots, \hat\beta_p) = \hat\beta' X'\mathbf{y} \quad (df = p+1), \qquad SS(\hat\beta_0) = n\bar y^2 \quad (df = 1),$$

$$SS(\hat\beta_1, \ldots, \hat\beta_p \mid \hat\beta_0) = SS(\hat\beta_0, \ldots, \hat\beta_p) - SS(\hat\beta_0) = SSR \quad (df = p).$$

If a submodel uses only $x_1, \ldots, x_q$ ($q < p$) with fit $\tilde\beta$, the extra sum of squares of $x_{q+1}, \ldots, x_p$ given $x_1, \ldots, x_q$ is

$$SS(\hat\beta_{q+1}, \ldots, \hat\beta_p \mid \hat\beta_0, \ldots, \hat\beta_q) = SS(\hat\beta) - SS(\tilde\beta) = SSR(\text{full}) - SSR(\text{sub}) = SSE(\text{sub}) - SSE(\text{full}),$$

with $p - q$ degrees of freedom. In matrix form, partition $\beta = (\beta_1', \beta_2')'$ with $\beta_1 = (\beta_0, \ldots, \beta_q)'$, $X = (X_1, X_2)$, and let $H_1 = X_1(X_1'X_1)^{-1}X_1'$, $H = X(X'X)^{-1}X'$:

$$SS(\beta_2 \mid \beta_1) = \mathbf{y}'(H - H_1)\mathbf{y}.$$

Theorem 8.1

(a) $H - H_1$ is symmetric idempotent of rank $p - q$, and $(H - H_1)(I - H) = 0$.

(b) Under normal errors, $SS(\beta_2 \mid \beta_1)/\sigma^2 \sim \chi^2(p-q, \lambda)$ with $\lambda = \beta_2' X_2'(I - H_1)X_2\beta_2/(2\sigma^2)$, independent of SSE.

(c) Hence, under $H_0 : \beta_{q+1} = \cdots = \beta_p = 0$,

$$F_0 = \frac{SS(\hat\beta_{q+1}, \ldots, \hat\beta_p \mid \hat\beta_0, \ldots, \hat\beta_q)/(p-q)}{MSE} \sim F(p-q, n-p-1).$$

Proof. (a) $HX = X$ gives $HX_1 = X_1$, hence $HH_1 = H_1$ and (transposing) $H_1H = H_1$. Then $(H-H_1)^2 = H - H_1 - H_1 + H_1 = H - H_1$, and $\operatorname{tr}(H - H_1) = (p+1) - (q+1)$. Also $(H - H_1)(I - H) = H - H - H_1 + H_1 = 0$.

(b) By Theorem 5.5(b) the noncentrality is $\frac{1}{2\sigma^2}(X\beta)'(H - H_1)X\beta$. Since $(H - H_1)X_1 = 0$ and $(H - H_1)X_2 = (I - H_1)X_2$, this equals $\beta_2' X_2'(I - H_1)X_2\beta_2/(2\sigma^2)$. Independence follows from (a) and Theorem 5.6.

(c) Under H_0, $\lambda = 0$. ■

This is the same test as the general linear hypothesis with $C = (0, I_{p-q})$ (Chapter 7), since $SSE(\text{sub}) - SSE(\text{full}) = Q$.

When the intercept is included in the regression sum of squares, one uses the uncorrected table: $SSR = \hat\beta' X'\mathbf{y}$ with $p+1$ degrees of freedom and $SST = \mathbf{y}'\mathbf{y}$ with n; then $MSR/MSE \sim F(p+1, n-p-1)$ tests $\beta = \mathbf{0}$.

8.2 Partial and sequential F tests

Partial F test. The contribution of a single regressor x_j given all others is $SS(\hat\beta_j \mid \text{all others})$, with one degree of freedom. The test of $H_0 : \beta_j = 0$,

$$F_0 = \frac{SS(\hat\beta_j \mid \hat\beta_0, \ldots, \hat\beta_{j-1}, \hat\beta_{j+1}, \ldots, \hat\beta_p)}{MSE} \sim F(1, n-p-1),$$

is the **partial F test**. It is equivalent to the t test, since

$$SS(\hat\beta_j \mid \text{others}) = \frac{\hat\beta_j^2}{c_{jj}}, \qquad F_0 = \Big(\frac{\hat\beta_j}{\sqrt{c_{jj}MSE}}\Big)^2 = t_0^2.$$

(This follows from Theorem 7.3 with $C = \mathbf{e}_j'$: $Q = \hat\beta_j^2/c_{jj}$.) A non-significant partial F means that x_j adds little *once the other variables are in the model*; it does not mean that x_j is unrelated to y. With correlated regressors, every partial F can be small while the overall F is large.

Sequential F tests. Enter the regressors one at a time in a chosen order, and at each stage test the newly added variable given those already in:

$$SS(\hat\beta_1 \mid \hat\beta_0), \quad SS(\hat\beta_2 \mid \hat\beta_0, \hat\beta_1), \quad \ldots, \quad SS(\hat\beta_p \mid \hat\beta_0, \ldots, \hat\beta_{p-1}).$$

These sum to SSR, and each is compared with the MSE of the model at that stage (or of the full model). The decomposition depends on the order unless the centred regressors are mutually orthogonal. The tests are valid only if the MSE used is unbiased for σ^2, which requires that the omitted variables have no effect; otherwise the denominator is inflated.

8.3 Joint confidence regions and simultaneous intervals

Motivation. Several individual 95% intervals do not hold simultaneously with probability 95%. Joint regions and Bonferroni or Scheffé intervals control the error rate for all statements at once.

Theorem 8.2 (Joint confidence region)

Under normal errors,

$$\frac{(\hat\beta-\beta)'X'X(\hat\beta-\beta)}{(p+1)MSE} \sim F(p+1, n-p-1),$$

so $\{\beta : (\hat\beta-\beta)'X'X(\hat\beta-\beta) \le (p+1)MSE\, F_\alpha(p+1,n-p-1)\}$ is a $100(1-\alpha)\%$ confidence ellipsoid for β.

Proof. $\hat\beta - \beta = (X'X)^{-1}X'\epsilon$, so $(\hat\beta-\beta)'X'X(\hat\beta-\beta) = \epsilon' H \epsilon$, and $\epsilon' H\epsilon/\sigma^2 \sim \chi^2(p+1)$. It is independent of $SSE = \epsilon'(I-H)\epsilon$, since $H(I-H)=0$. ∎

Separate 95% intervals for $\beta_0, \ldots, \beta_p$ do not have joint coverage 95%. Two remedies:

Theorem 8.3 (Bonferroni and Scheffé intervals)

(a) (Bonferroni) The intervals $\hat\beta_j \pm t_{\alpha/(2(p+1))}(n-p-1)\sqrt{c_{jj}MSE}$, $j = 0, \ldots, p$, cover all β_j simultaneously with probability at least $1-\alpha$.

(b) (Scheffé) With probability $1-\alpha$, simultaneously for all $\mathbf{a} \in \mathbb{R}^{p+1}$,

$$\mathbf{a}'\beta \in \mathbf{a}'\hat\beta \pm \sqrt{(p+1)F_\alpha(p+1,n-p-1)}\sqrt{\mathbf{a}'(X'X)^{-1}\mathbf{a}\, MSE}.$$

Proof. (a) If A_j is the event that the j-th interval covers, $P(A_j^c) = \alpha/(p+1)$, and $P(\bigcap A_j) \ge 1 - \sum P(A_j^c) = 1-\alpha$.

(b) By the Cauchy–Schwarz inequality in the inner product defined by $(X'X)^{-1}$,

$$\max_{\mathbf{a}\neq\mathbf{0}} \frac{\{\mathbf{a}'(\hat\beta-\beta)\}^2}{\mathbf{a}'(X'X)^{-1}\mathbf{a}} = (\hat\beta-\beta)'X'X(\hat\beta-\beta),$$

so the event in (b) is exactly the joint confidence region of Theorem 8.2. ■

Taking $\mathbf{a} = \mathbf{x} = (1, x_1, \ldots, x_p)'$ in (b) gives a **confidence band** for the whole regression surface, valid for all $\mathbf{x}$ at once. For a few coefficients Bonferroni intervals are shorter; Scheffé intervals pay for covering infinitely many combinations.

8.4 Testing equality of regression models

Two populations follow

$$\mathbf{y}_1 = X_1\beta_1 + \epsilon_1, \qquad \mathbf{y}_2 = X_2\beta_2 + \epsilon_2,$$

with n_1 and n_2 observations, design matrices of size $n_i \times (p+1)$, and independent $N(\mathbf{0}, \sigma^2 I)$ errors with a *common* σ^2. To test $H_0 : \beta_1 = \beta_2$:

(1) Full model: fit each separately, $SSE(F) = SSE_1 + SSE_2$, with $n - 2(p+1)$ degrees of freedom ($n = n_1 + n_2$).

(2) Reduced model: stack $\mathbf{y} = (\mathbf{y}_1', \mathbf{y}_2')'$ and $X = (X_1', X_2')'$ and fit one β_0: $SSE(R)$, with $n-(p+1)$ degrees of freedom.

(3) Under H_0,

$$F_0 = \frac{\{SSE(R) - SSE(F)\}/(p+1)}{SSE(F)/(n-2p-2)} \sim F(p+1, n-2p-2).$$

The reduced model is the full model (with block-diagonal design $\mathrm{diag}(X_1, X_2)$ and parameter $(\beta_1', \beta_2')'$) under the $p+1$ linear restrictions $\beta_1 - \beta_2 = \mathbf{0}$, so this is again a general linear hypothesis. For k groups and $H_0 : \beta_1 = \cdots = \beta_k$, the numerator has $(k-1)(p+1)$ degrees of freedom and the denominator $n - k(p+1)$.

The test assumes equal error variances; with $k = 2$ this can be checked first by $F = MSE_1/MSE_2 \sim F(n_1 - p - 1, n_2 - p - 1)$.